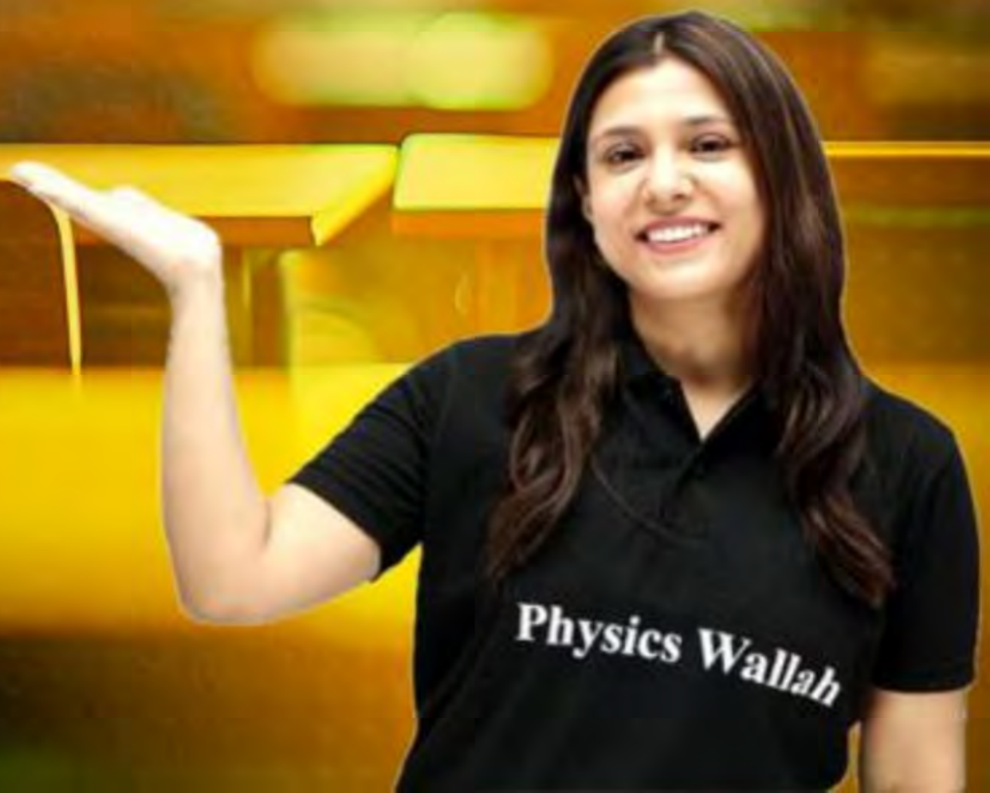
RADIANT 2026

Biology

Five Kingdoms Classification

Lecture - 02

By - Upasana Mongia Ma'am



Topics *to be covered*

1. Moncra
2. Protista
3. fungi
4. Plantae.
5.



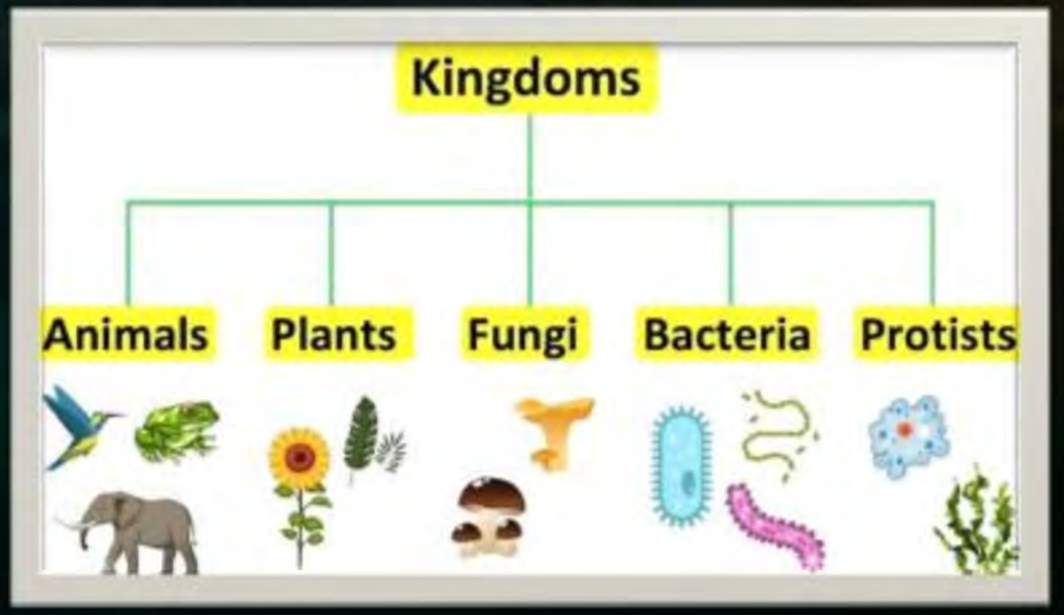






Introduction of Five Kingdoms Classification

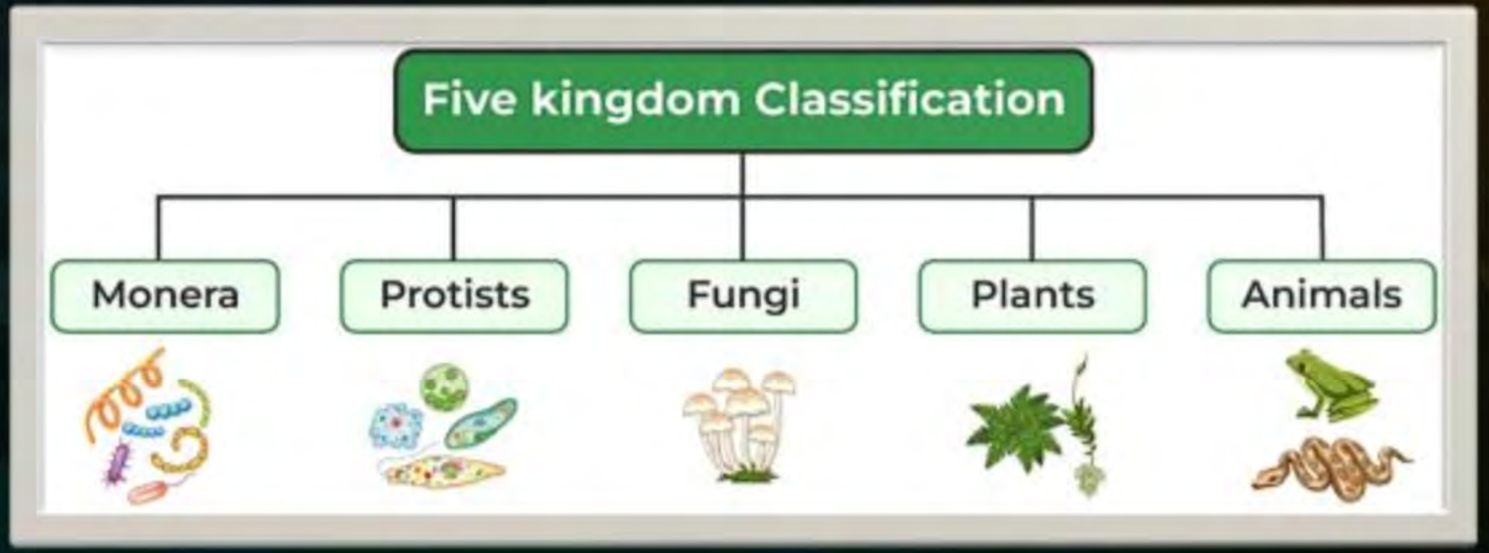
- Based on their characteristics like cell structure, body organization, mode of nutrition, source of nutrition, reproduction, and interrelationship with others.
- This system was developed by Robert H. Whittaker in 1969.



Introduction of Five Kingdoms Classification

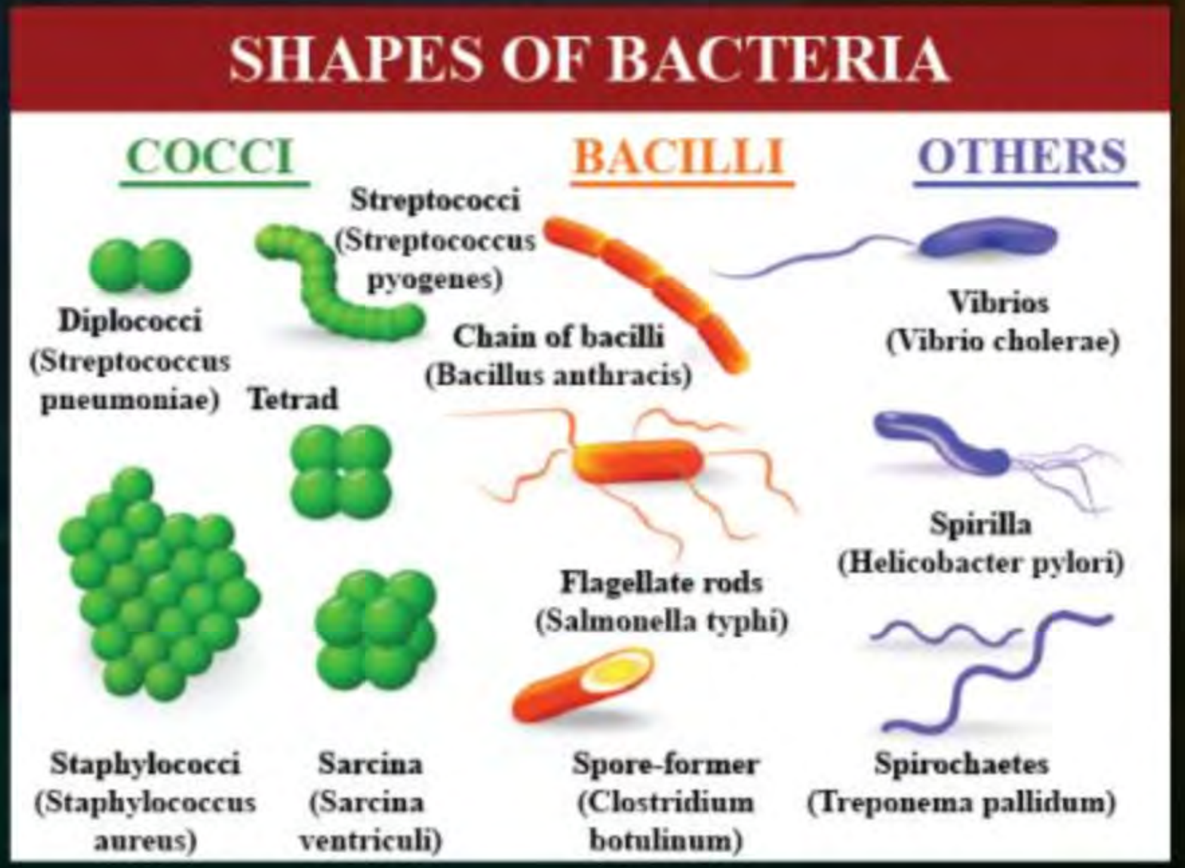
- According to the five-kingdom classification, a living being can be sub-categorised into five major kingdoms:

  - Kingdom Monera
  - Kingdom Protista
  - Kingdom Fungi
  - Kingdom Plantae
  - Kingdom Animalia



Types of Monerans

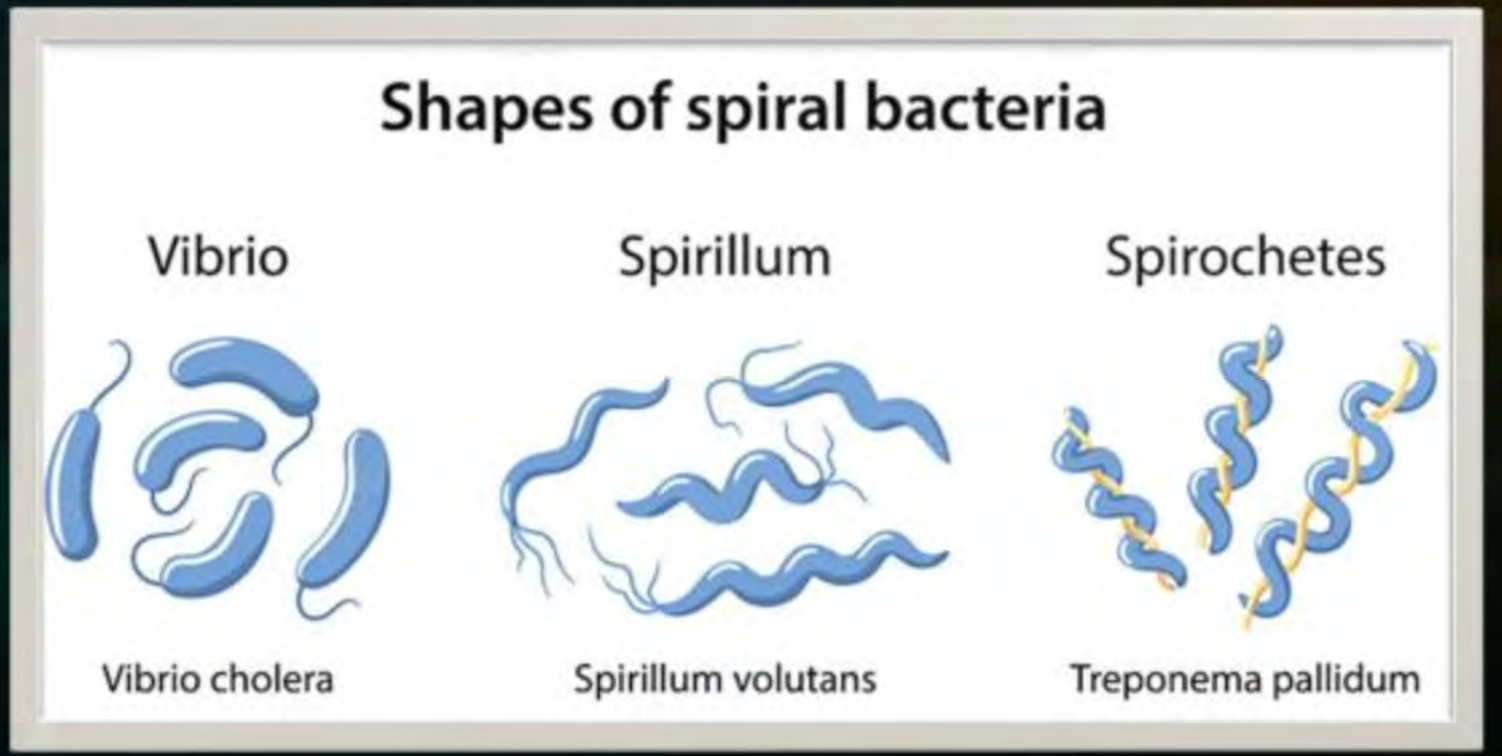
- They can be categorized into four different kinds based on the shape they form. These are:
- Coccus / Cocci- These bacteria are known to have a spherical shape.
- Bacillus / Bacilli- These bacteria are known to have the shape of a rod.



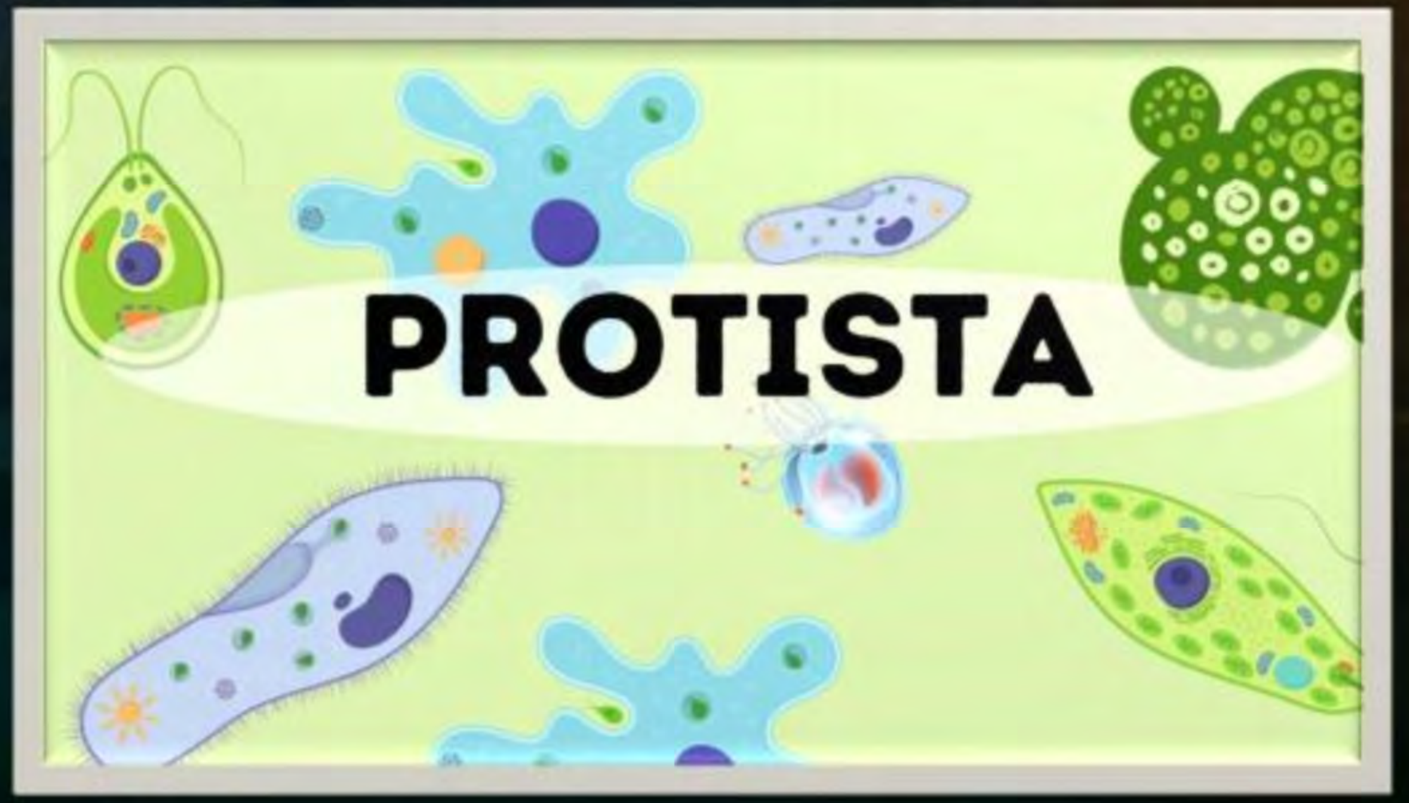
Types of Monerans

- Vibram / Vibrio- These bacteria are known to form the shape of a comma.
- Spirillum / Spirilla- These bacteria are spiral-shaped.





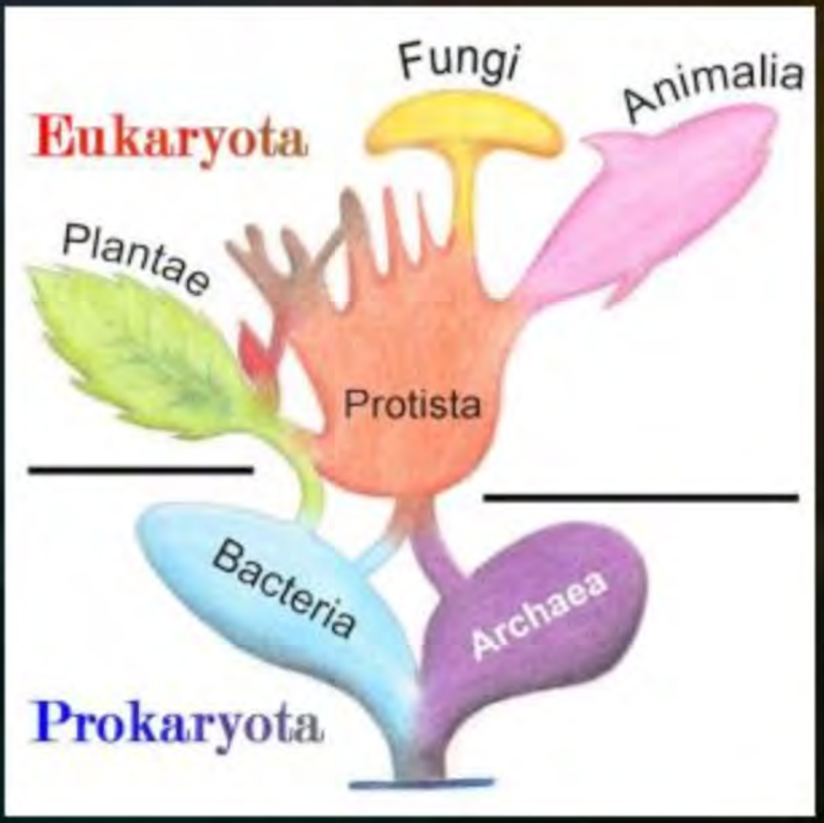
Kingdom Protista/Protoctista



Features of Kingdom Protista

- Organisms grouped under Kingdom Protista are unicellular and eukaryotic.
- Sometimes they are simple multicellular organisms.
- They include unicellular autotrophic organisms also non green heterotrophic organisms like *Euglena, Amoeba,Paramecium)*

Heterotrophic



Euglena is ——.

A) Heterotrophic ✓

B) Autotrophic ✓

C) Both A & B

D) None ——.



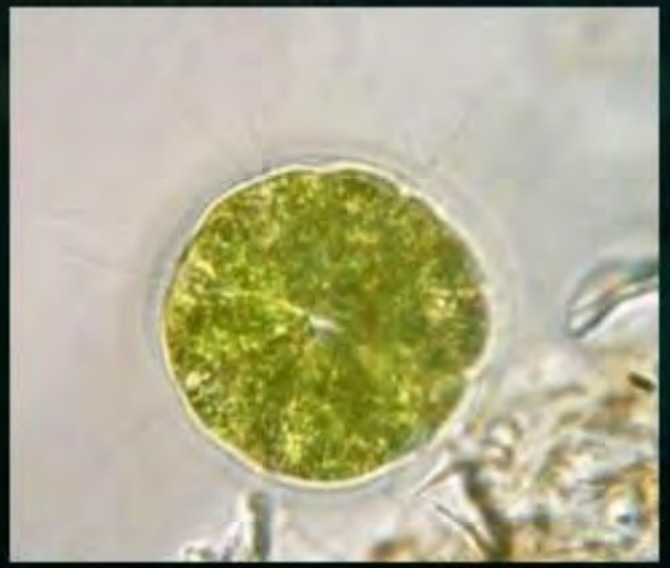


Features of Kingdom Protista

- Some multicellular photosynthetic organisms like *Pandorina*
- Some organisms also have appendages that are also called flagella or cilia. They use these mechanisms to move around.
- They opt for sexual or asexual modes of reproduction.



Paramoecium





Features of Kingdom Protista

- Protists can have two major groups- the first bear similarity with animals cells in the sense that they do not have cell walls; the second bears similarity to the plant cells which have cell walls and have characteristics similar to that of algae.

Kingdom Fungi

- Kingdom Fungi consists of eukaryotic, heterotrophic, multicellular organisms.

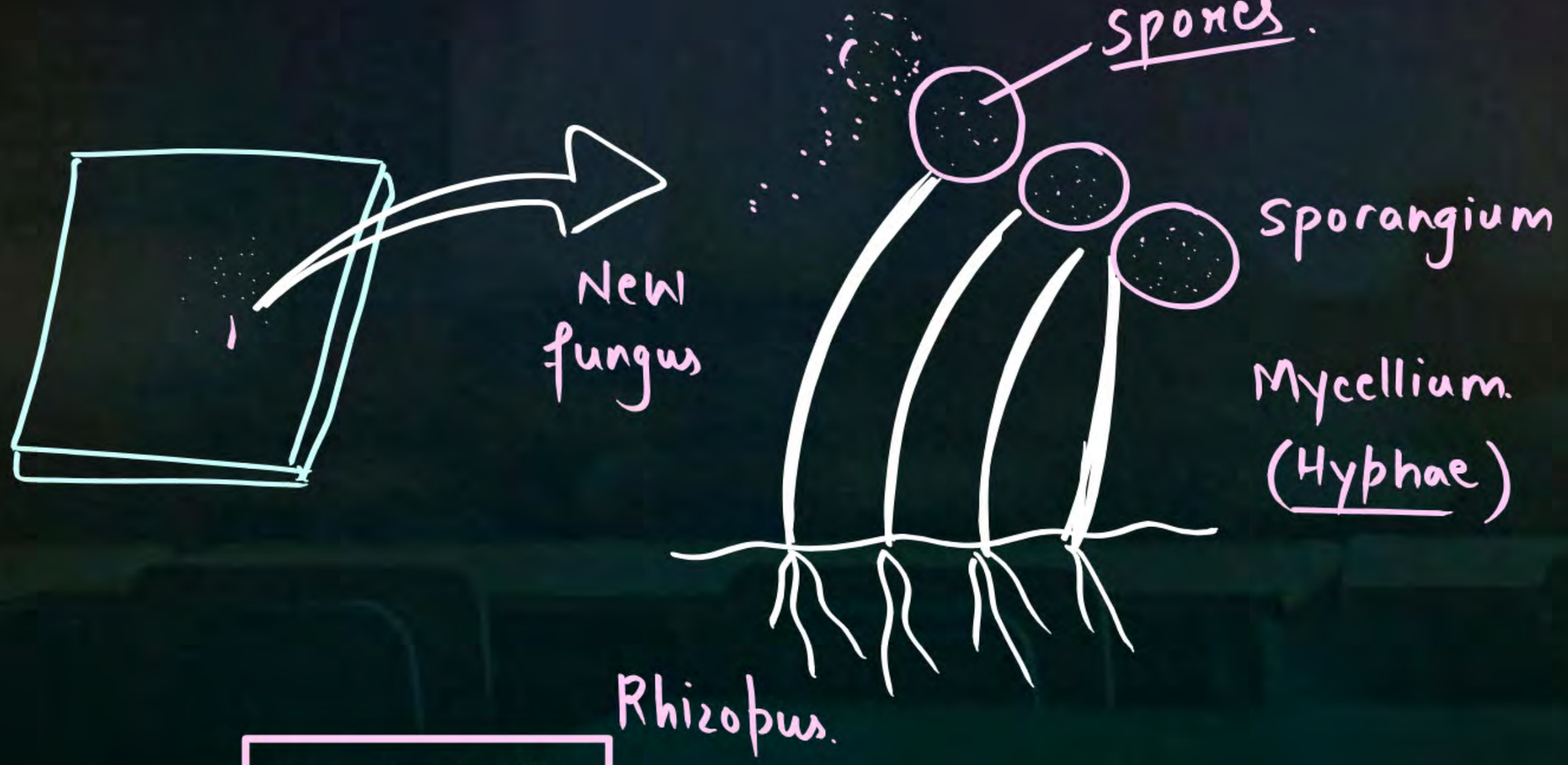
Monera → Bacteria
- Prokaryotic
- unicellular

Protista
- Eukaryotic
- unicellular

Q → Smallest ?
→ Without Cell Wall ?

PPLO / Mycoplasma



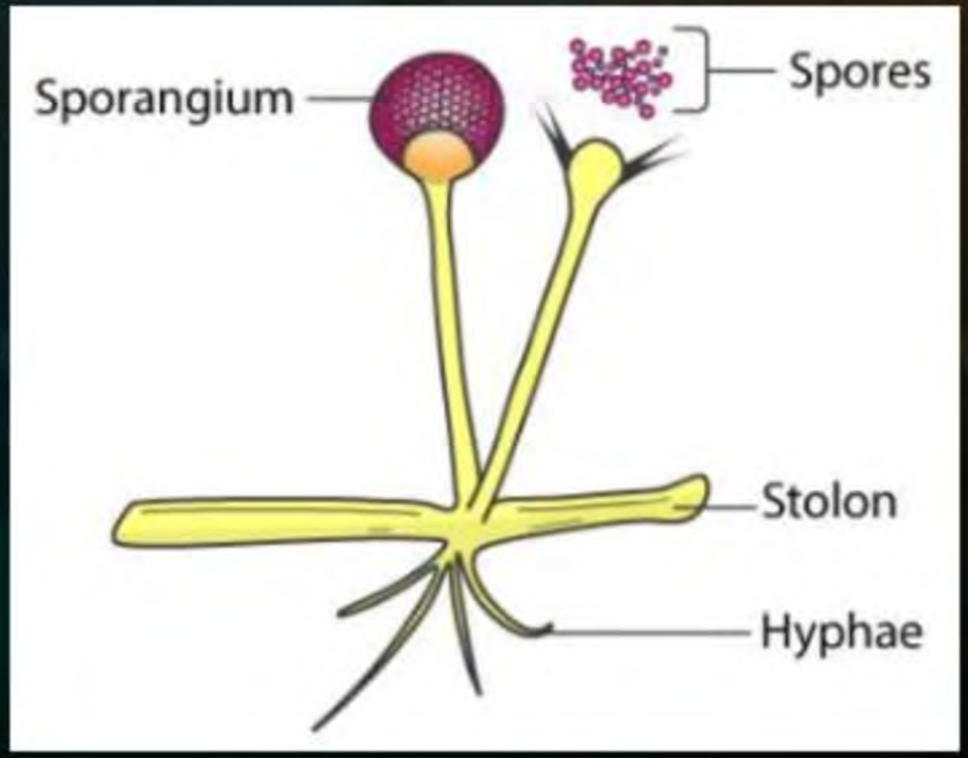


Bread Mould

Features of Kingdom Fungi

- Fungi can be either unicellular or multicellular; yeast is an example of unicellular and mushrooms are an example of multicellular.
- They are heterotrophic organisms.
- They are also Saprophytic
- They can also often be parasitic.

- Their cell wall is made up CHITIN. ★



The organism which take food from dead & decaying matter.

A) Parasitic ⟶ Host body

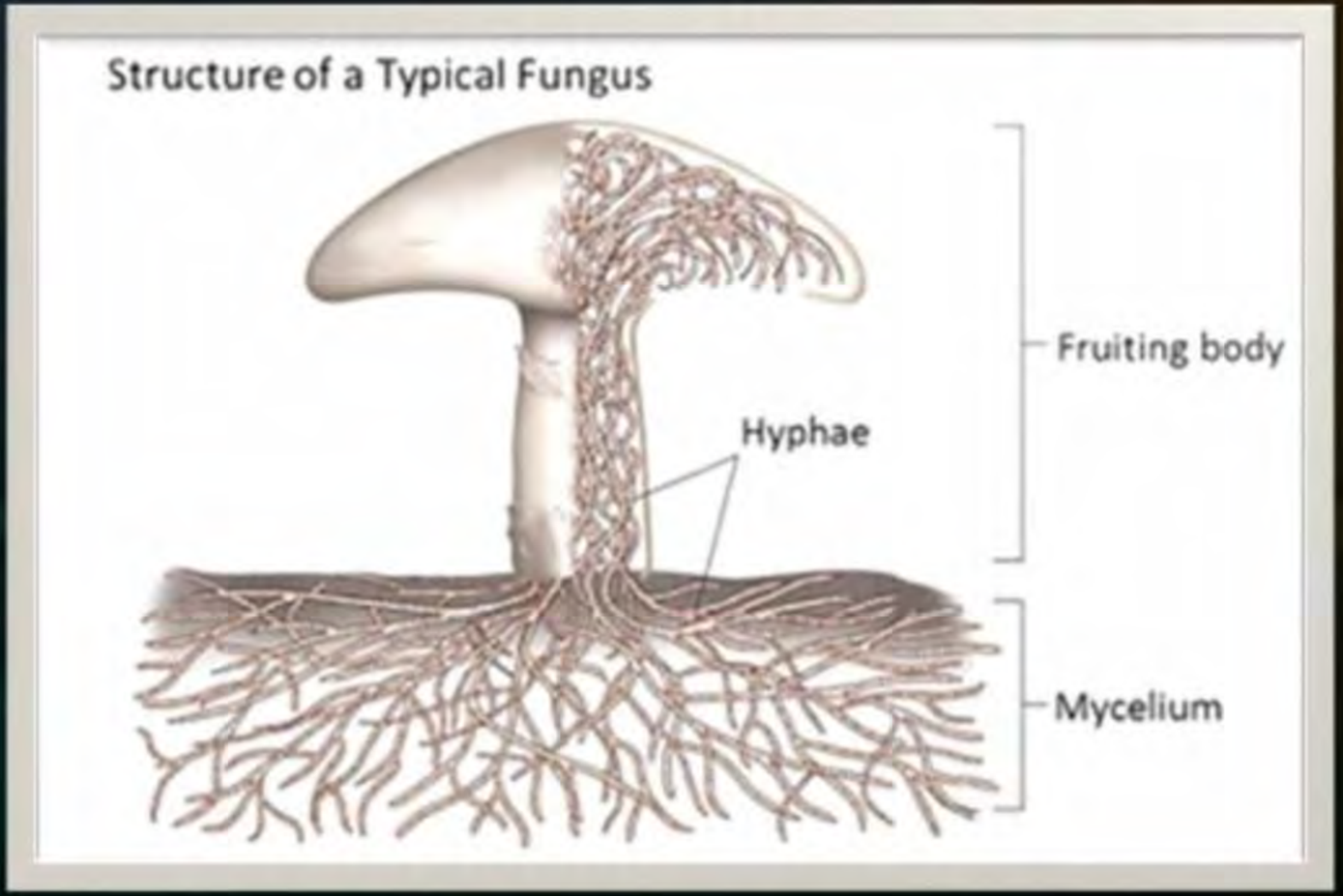
B) Saprophytic ✓

Features of Kingdom Fungi

- Reproduction occurs through both sexual and asexual methods of spore formation.

Features of Kingdom Fungi

- Examples of fungi include mushrooms, yeast, and aspergillus.
- Fungi are incapable of locomotion.
- They consist of fine threads known as hyphae. An entire web of hyphae is known as mycelium.

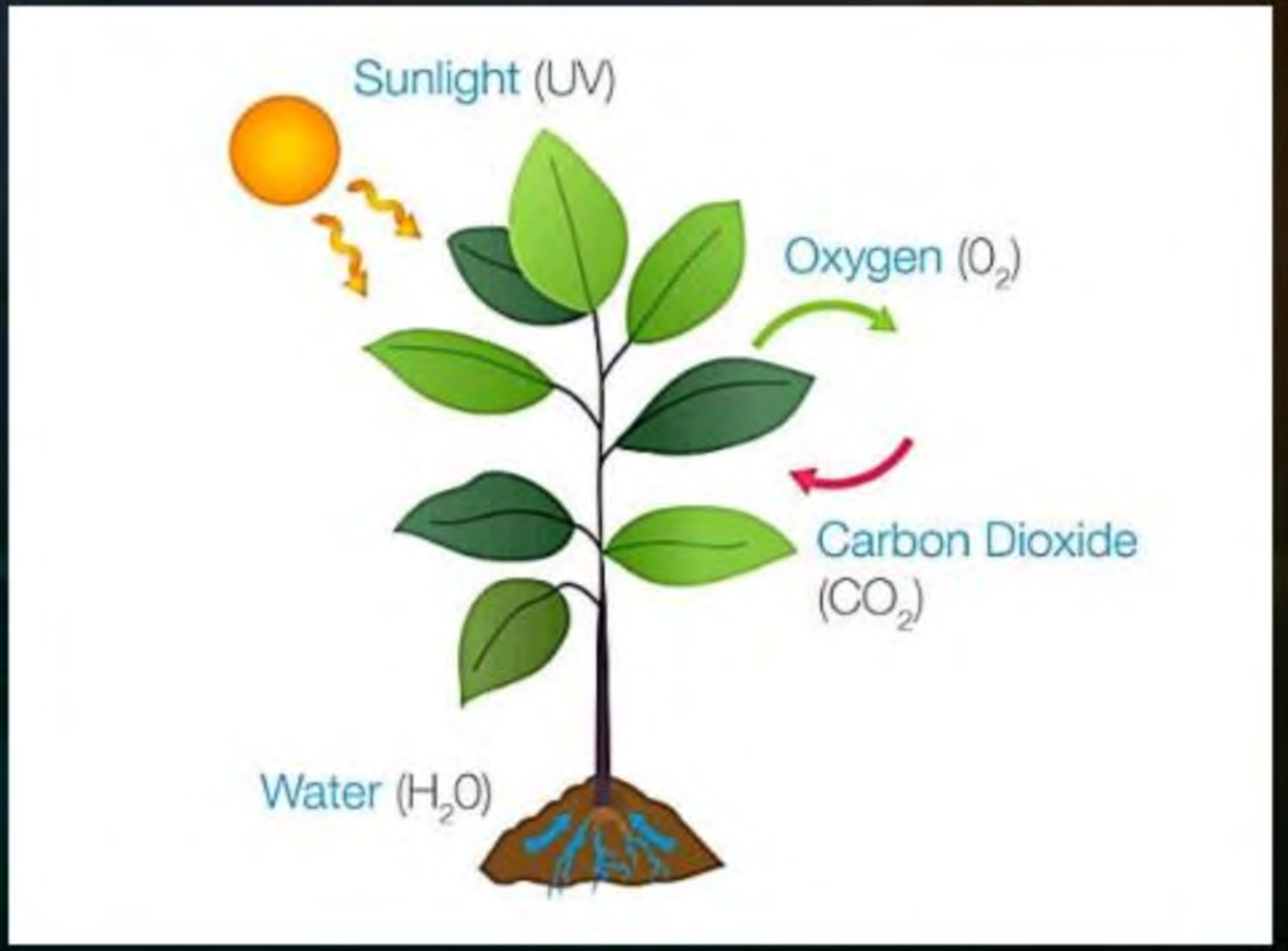


Kingdom Plantae



Features of Kingdom Plantae

- These organisms are all eukaryotic organisms and have chloroplasts.
- They are multicellular.
- They have a cell wall that is highly distinct and made of cellulose.



Don't write

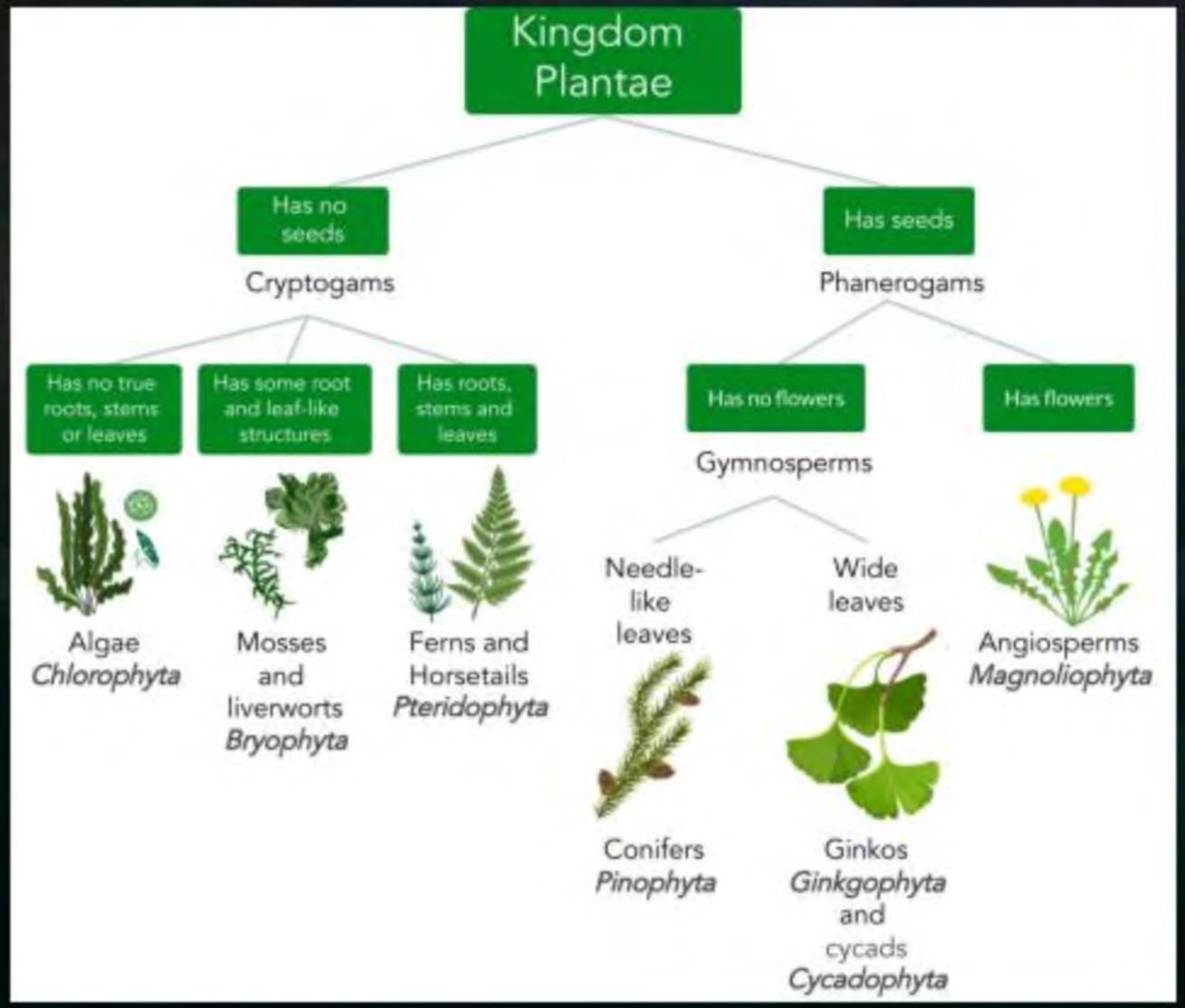


Kingdom Plantae

↓ 5 Division

Thallophyta | Bryophyta | Pteridophyta | Gymnospermae ✓ | Angiospermae ✓

Kingdom : Plantae







Division 1 : Thallophyta

❑ Following are its basic features:

- ➢ Basic and elementary plants with undifferentiated body parts. Have thallus like body.

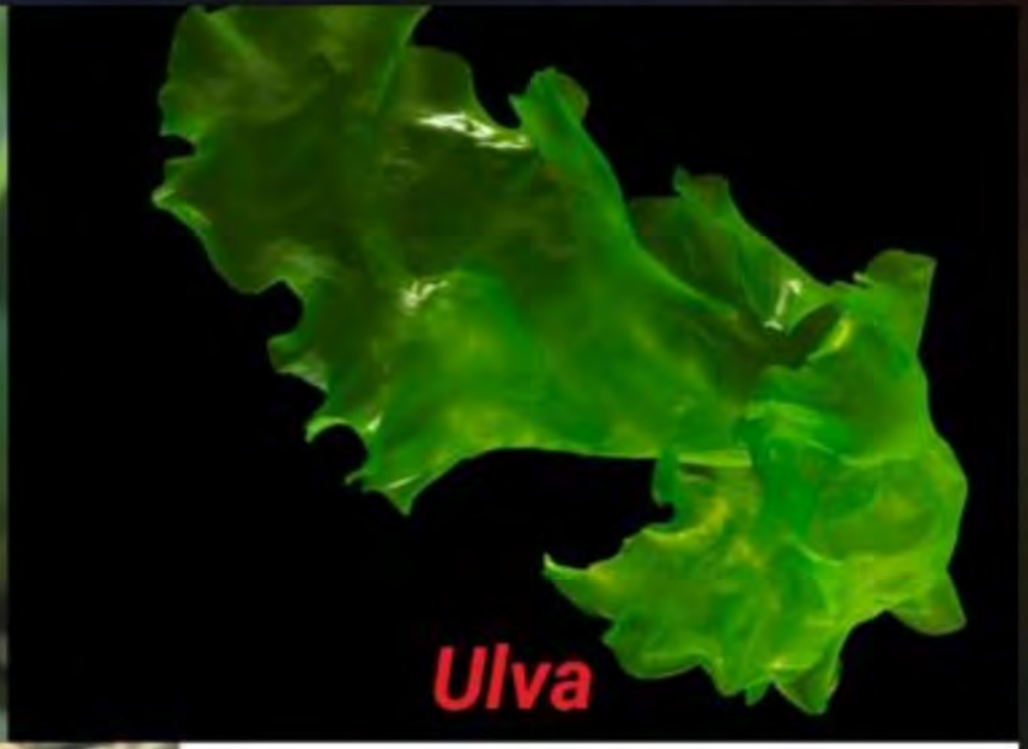
(No root, stem, leaf).

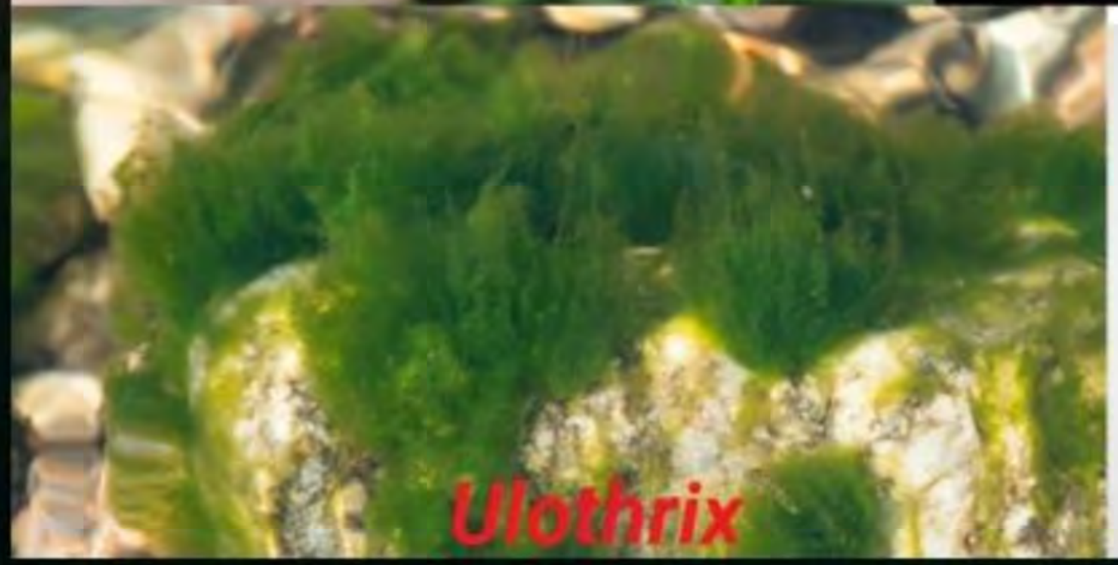


- Generally called algae.
- Mainly found in water(aquatic)
- Example- Ulva, Spirogyra, Ulothrix, Cladophora, Chara, seaweed ,kelps

- Asexual Reproduction → Spores, fragments


Spirogyra


Ulva


Ulothrix


Cladophora

Division 2: Bryophyte

Following are its basic features:

- Body structure differentiated but not fully developed,autotrophic
- Possess root like(rhizoids) ,stem like(caulid)and leaf like (phyllids).
- Reproduce through spores. (Asexual Repⁿ)
- Found on both land and water therefore known as 'Amphibians of Plantae kingdom'.
- Example – liwerwort (Marchantia, Riccia), mosses(Funaria), hornwort


Riccia
Anthoceros
Funaria
Marchantia



Division 3: Pteridophyta

Following are its basic features:

- Differentiated body structure- leaves, stems, roots, etc.
- Reproduce through spores underside
- Examples-Marsilea, fern, horsetails
- Vascular bodies might be present.

Marsilea



Ferns



Horsetail



Division 4: Gymnosperms

- **Following are its basic features:**
- Differentiated body parts
- Has reproductive structures called cones which bear male pollen grains and female ovule
- Naked seeds without fruits or flowers
- Two types- cycads and conifers
- Examples- Pines(deodar), Cycus, Ginkgo,fir

[cycad] →



[Conifers] →



Division 5: Angiosperms

- **Following are its basic features:**
- Also known as Flower - bearing plants.
- Later on flower becomes fruit.
- Seeds are inside the fruit. → covered seeds
- Divided into Monocots (maize, rice, grass) and Dicots (pea , potato,apple,sunflower,rose)



MONOCOTS



1 cotyledon



fibrous roots



parallel leaf veins



scattered stem vascular bundles

xylem & phloem

DICOTS



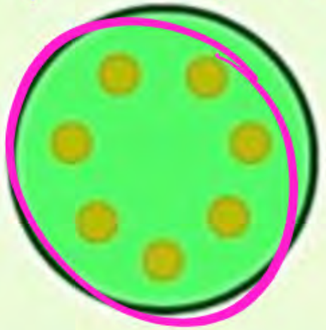
2 cotyledons



tap root



Reticulate net leaf veins



vascular bundles form ring

Question

A plant that has seeds but no flowers and fruits?

A Bryophytes spores

B Gymnosperms

C Mosses spores

D Pteridophytes spores / No seeds

Question

Double fertilization is the characteristic of

flowers

A Algae

B Gymnosperms

C Fungi

D Angiosperms

Question

The cell wall of Spirogyra contains

plantae

A Cellulose

B Chitin → fungus

C Lignin

D Suberin


Thank You